Power BI Power Query – Advanced Data Transformation

1. Introduction to Ongoing Data Transformation

In Power BI, data transformation is a critical preparatory step for building accurate and meaningful visual reports. Power Query Editor offers a wide range of transformation features that help clean, modify, enrich, and shape your data before loading it into the Power BI data model.

In the previous steps, various transformations—such as removing errors, changing data types, renaming columns, and rounding decimal values—were already applied to the product dataset. In this session, we continue exploring advanced transformations like subtraction, division, percentage formatting, duplicate vs reference tables, and grouping operations.

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2. Mathematical Operations in Power Query (Add Column Tab)

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Power Query provides standard mathematical operations under the **Add Column** tab. These operations help create new calculated fields without modifying existing data. They include addition, subtraction, multiplication, division, rounding, and more.

We explore two critical operations: **Subtraction** and **Divide**.

2.1 Subtraction Operation — To Calculate Profit

Concept

Profit = Selling Price – Cost Price
Using subtraction, we can derive the profit for each product item.

Step-by-Step Process

1. Select the columns:
    - **Selling price column** (e.g., Item S-Price)
    - **Cost price column**

2. Go to the **Add Column** tab.
3. In the ribbon, choose **Standard > Subtract**.
4. Power Query automatically creates a new column with the difference between the two values.

Example

For the first item:
Selling Price = 48.99
Cost Price = 18.32
Profit = 48.99 – 18.32 = **30.67**

Post-Processing

- Rename the new column to **Profit** by double-clicking the header.
- Apply rounding: Change decimal places to **2**, ensuring uniform formatting.

This step completes the profit calculation for each product in the dataset.

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3. Division Operation – To Calculate Profit Margin

Profit margin is an essential metric showing how much percentage of the selling price is profit.

Formula

Profit Margin = Profit ÷ Selling Price

Steps to Create Profit Margin Column

1. Select **Profit** column.
2. Select the Selling **Price** column.
3. In the **Add Column** tab → choose **Standard > Divide**.
4. Choose the default **Divide** function (NOT Divide Integer), since we want the exact decimal ratio.

Example

Profit = 30.67
Selling Price = 48.99
Profit Margin = 30.67 ÷ 48.99 = 0.6260 (≈ 62.60%)

Renaming and Formatting

- Rename the column to **Profit Margin**.
- Apply **Percentage** formatting.
- Power Query automatically adjusts decimal places when converting to percentage (usually to 2 decimals).

Important Note

If decimal rounding is applied before percentage formatting, the result may differ.
Correct sequence:

1. Convert to **Percentage**
2. Allow Power Query to apply decimal formatting automatically

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4. Managing Tables in Power Query: Delete, Duplicate & Reference

Inside Power Query Editor, multiple tables may exist, and you may want to manage them for analytical or transformation purposes. Under the **Home** tab, the **Manage** dropdown provides:

- **Delete Table**
- **Duplicate Table**
- **Reference Table**

Let's understand each one in detail.

4.1 Delete Table

Used to remove a table from the Power Query environment.

Ways to Delete

- Right-click the table name → **Delete**
- Home tab → Manage → **Delete**

Useful for removing unnecessary or outdated data sources.

4.2 Duplicate Table

What is a Duplicate Table?

Creates an *exact* copy of the selected table:

- All columns
- All transformed steps
- All data values

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Key Characteristics

- It is **independent** of the original table.
- Any future changes in the original table **do not affect** the duplicate.
- Useful for experimenting with transformations without touching the primary dataset.

How to Create

Right-click table → **Duplicate**
or
Home → Manage → **Duplicate**

Use Case

Testing advanced transformations or trial-and-error operations without risking the integrity of the primary table.

4.3 Reference Table

What is a Reference Table?

A Reference Table is **linked** to the original table and pulls data dynamically from it.

Characteristics

- Contains the **same columns** as the original.
- Displays **almost no applied steps** (only Source).
- Any change in the original table **automatically updates** the reference table.

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Difference from Duplicate

Feature	Duplicate Table	Reference Table
Copies Steps	Yes	No
Linked to Original	No	Yes
Updates when Original Changes	N0	Yes
Use Case	Experiments	Summaries, grouping, lookups

How to Create

Right-click table → **Reference**
or
Home → Manage → **Reference**

Why Use Reference Tables?

Best suited for:

- Summary tables
- Aggregations
- Group By operations
- Data modeling with parent-child relationships

Reference tables reduce redundancy and ensure data consistency.

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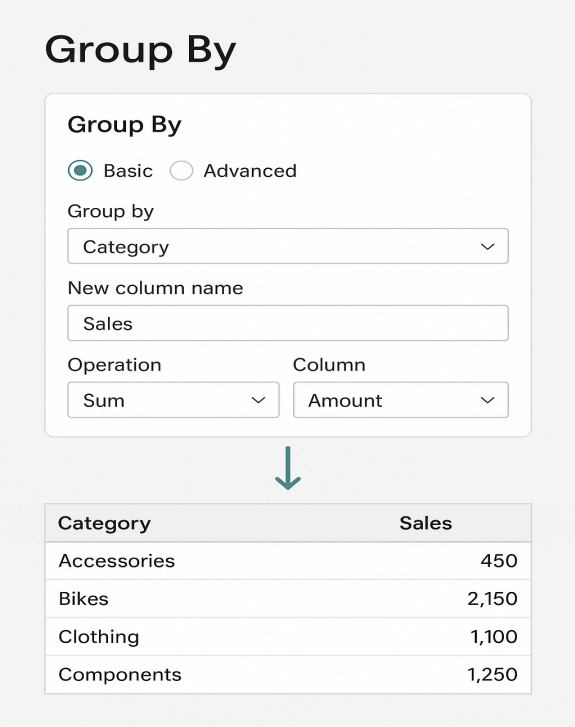
5. Group By Operation in Power Query

Grouping helps summarize data by grouping rows based on one or more column values. Power Query offers two modes:

- **Basic Group By**
- **Advanced Group By**

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Group By is commonly used to create summary tables derived from detailed datasets.



5.1 Basic Group By Operation

Purpose

To aggregate data based on a single column, using operations like:

- Count Rows
- Sum
- Average
- Maximum
- Minimum

Example Scenario

You want to know how many products are associated with each **ModelName**.

Steps

1. Select the **ModelName** column.
2. Choose **Home tab → Group By**.
3. Choose:
    - Column to group by: **ModelName**
    - New column name: e.g., **Number of Items**
    - Operation: **Count Rows**
4. Click **OK**.

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Output

A summary table showing:

Model Name	Count
ROAD650	12
MOUNTAIN300	8
etc.	

Sorting

Sort descending to find top-selling models and least-selling ones.

5.2 Advanced Group By Operation (Multiple Columns)

Purpose

Allows grouping based on multiple columns such as:

- Model Name
- Size
- Category
- Region
- Date Hierarchy (Year, Month)

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Example

Group products by:

1. **Model Name**
2. **Size**

To get a more detailed breakdown.

Steps

1. Create a new Reference Table.
2. Go to **Home → Group By**.
3. Choose **Advanced mode**.
4. Add two grouping levels:
    - ModelName
    - Size
5. For operation:
    - New column name: **Number of Products**
    - Operation: **Count Rows**
6. Click **OK**.

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Output Example

ModelName	Size	Number of Products
ROAD650	48	2
ROAD650	58	2
ROAD650	XXS	3

etc.

Uses

- Multi-level summaries
- Product/category analysis
- Inventory breakdown
- Hierarchical reporting

Since it works on a *reference table*, any changes in the main list of products will automatically update the summary table.

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6. Practical Use Cases of Group By and Reference Tables

Why Reference Tables + Group By are powerful

They help create dynamic aggregate tables that:

- Auto-refresh with source data
- Require no repeated transformations
- Avoid duplication of information

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Common Use Cases

1. Sales summaries by product model
2. Profitability analysis by category
3. Inventory count by size
4. Customer segmentation
5. Regional sales summaries
6. Creating dimension tables for star schema

7. Understanding the Dependency Relationship: Duplicate vs Reference

Duplicate Table = Independent Copy

- Best for testing or branching transformations.
- Does not inherit future updates from the parent table.

Reference Table = Dependent (Child) Table

- Ideal for creating summary tables.
- Always reflects the parent table's latest shape and transformations.

Visualization of Relationship



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8. Finalizing the Data Transformation

After completing:

- Profit calculation
- Profit margin calculation
- Percentage conversion
- Duplicate and reference table setups
- Group By summaries

…the final step is to load the transformed data into Power BI.

Closing Steps

Click **Close & Apply** on the Home tab.

Power BI will:

- Validate applied steps
- Refresh data model
- Load tables into the report view

This finalizes the transformation phase for the dataset.

9. Summary of Key Learning Concepts

Mathematical Operations

- Subtraction: Used for Profit calculation
- Division: Used for Profit Margin
- Formatting: Percentage formatting & rounding

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Table Management

Feature	Purpose
Delete	Remove unused tables
Duplicate	Independent copy for trial transformations
Reference	Linked dependent table for summaries

Group By

- Basic Grouping: single-level aggregation
- Advanced Grouping: multi-level aggregation
- Reference tables recommended for summarization

10. Conclusion

This session explored essential data transformation features in Power Query that are critical for creating meaningful and accurate Power BI reports. Understanding subtraction, division, and formatting helps automate financial calculations. Learning duplicate and reference tables teaches proper data architecture inside Power Query. Finally, mastering Group By empowers analysts to generate powerful summary tables without altering primary datasets.

Together, these skills elevate your Power BI data preparation abilities, enabling cleaner data models, more efficient analytical workflows, and automation-ready reporting systems.

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